ALLIANCE UNIVERSITY

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# REPORT

# On

**SUBJECT**

MACROECONOMICS AND BUSINESS POLICY ENVIRONMENT


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## Correlation Study: GDP and Human Development

## 1. Introduction

Economic growth has traditionally been evaluated using Gross Domestic Product (GDP). However, modern development economics argues that growth should also improve people's well-being. Indicators such as the Human Development Index (HDI) measure health, education, and living standards.

This study therefore examines whether higher GDP is actually associated with better human development outcomes.

The research attempts to statistically verify the idea proposed by welfare economists — that economic growth should translate into improvements in quality of life.

## 2. Need for Correlation Analysis

There is a major debate in economics:

- One view: **Economic growth automatically improves living standards**
- Other view: **Growth and welfare may not move together**

Simply observing GDP and HDI separately does not show their relationship. Correlation analysis is needed because it helps us determine:

Whether increases in GDP are actually accompanied by improvements in human development.

In short, we are testing:
**Does growth lead to development?**

## 3. Significance of the Study

This analysis is important for three reasons:

### Policy relevance

Governments often target GDP growth. If correlation is weak, policies should instead focus on education, healthcare, and inequality.

### Academic relevance

It supports or challenges welfare economics theories (capability approach and beyond GDP framework).

### Practical relevance

Helps understand whether citizens truly benefit from economic expansion.

## 4. Variables Used in the Study

### Independent Variable (X)

**GDP (Gross Domestic Product)**
Represents economic output and national income.

### Dependent Variable (Y)

**HDI (Human Development Index)**
Composite index based on:

- life expectancy (health)
- education level
- standard of living

### Type of Data

- Time series data
- Country: India
- Period: multiple years (as per dataset used)

| India GDP & HDI Data (1990 –2023) | | | | |
|---|---|---|---|---|
| **Year** | **GDP (US$ Bn)** | **HDI (0 - 1)** | **GDP GROWTH %** | **HDI CHANGE** |
| 1990 | 326 | 0.434 | -15.95 | 0.004 |
| 1991 | 274 | 0.438 | 6.93 | 0.004 |
| 1992 | 293 | 0.442 | -3.07 | 0.005 |
| 1993 | 284 | 0.447 | 17.25 | 0.006 |
| 1994 | 333 | 0.453 | 9.91 | 0.008 |
| 1995 | 366 | 0.461 | 9.02 | 0.007 |
| 1996 | 399 | 0.468 | 6.02 | 0.008 |
| 1997 | 423 | 0.476 | 1.18 | 0.007 |
| 1998 | 428 | 0.483 | 8.88 | 0.008 |
| 1999 | 466 | 0.491 | 2.15 | 0.006 |
| 2000 | 476 | 0.497 | 3.78 | 0.008 |
| 2001 | 494 | 0.505 | 6.07 | 0.007 |
| 2002 | 524 | 0.512 | 17.94 | 0.009 |
| 2003 | 618 | 0.521 | 14.72 | 0.028 |
| 2004 | 709 | 0.549 | 15.66 | 0.007 |
| 2005 | 820 | 0.556 | 14.63 | 0.007 |
| 2006 | 940 | 0.563 | 28.72 | 0.008 |
| 2007 | 1210 | 0.571 | -1.65 | 0.008 |
| 2008 | 1190 | 0.579 | 12.61 | 0.007 |
| 2009 | 1340 | 0.586 | 24.63 | 0.009 |
| 2010 | 1670 | 0.595 | 8.98 | 0.008 |
| 2011 | 1820 | 0.603 | 0.38 | 0.004 |
| 2012 | 1827 | 0.607 | 1.59 | 0.005 |
| 2013 | 1856 | 0.612 | 9.86 | 0.012 |
| 2014 | 2039 | 0.624 | 3.19 | 0.007 |
| 2015 | 2104 | 0.631 | 9.08 | 0.009 |
| 2016 | 2295 | 0.64 | 15.51 | 0.005 |
| 2017 | 2651 | 0.645 | 1.96 | 0.002 |
| 2018 | 2703 | 0.647 | 6.18 | 0.004 |
| 2019 | 2870 | 0.651 | -7.07 | -0.006 |
| 2020 | 2667 | 0.645 | 18.86 | -0.012 |
| 2021 | 3170 | 0.633 | 6.85 | 0.011 |
| 2022 | 3387 | 0.644 | 5.34 | 0.012 |
| 2023 | 3568 | 0.656 | NIL | -0.656 |

| **CORRELATION** | | |
|---|---|---|
| | *Column 1* | *Column 2* |
| Column 1 | 1 | |
| Column 2 | 0.198034 | 1 |

## 5. Procedure (Step-by-Step)

## Explain the Procedure (Actual Work Done)

After collecting the annual data of India's GDP and HDI and arranging it in a spreadsheet (Year | GDP | HDI), a correlation analysis was performed to measure the relationship between the two variables.

The following steps were followed:

1. GDP values were entered in one column and HDI values in another column.
2. The data range was selected in the spreadsheet software (Excel/Google Sheets).
3. The built-in correlation function was applied: **CORREL(GDP range, HDI range)**
4. The software automatically calculated the Pearson correlation coefficient (r), which measures the strength and direction of linear association between GDP and HDI.

No regression or scatter plot analysis was performed; only correlation analysis was used to examine the relationship.

## Results

The calculated correlation coefficient between GDP and HDI is:

**r = 0.198034**

Since the value of **r lies between 0 and +1**, it indicates a **positive relationship**, but the magnitude is **very low**.

Interpretation scale:

- 0.00 – 0.20 → very weak relationship
- 0.21 – 0.40 → weak
- 0.41 – 0.60 → moderate
- 0.61 – 0.80 → strong
- 0.81 – 1.00 → very strong

Thus, **0.198 ≈ very weak positive correlation**.

## Interpretation of Results

The result shows that increases in GDP are only slightly associated with improvements in HDI. In practical terms, economic growth in India does not automatically translate into proportional improvements in health, education, and living standards.

This implies:

- Growth alone is insufficient for human development.
- Other factors — public health infrastructure, education policy, inequality, and social welfare programs — play a major role in improving HDI.
- A country may experience rising national income while human development improves slowly.

The finding aligns with the human development approach: economic expansion is helpful but not enough; targeted social investment is necessary.

## Conclusion

The correlation analysis reveals a weak positive relationship between GDP and HDI in India ($r = 0.198$). While economic growth contributes to development, its impact on human well-being is limited when not supported by effective social policies. Therefore, GDP growth should not be treated as the sole indicator of progress.

For meaningful development, economic policy must combine growth with investments in healthcare, education, inclusion, and social protection so that national income expansion leads to real improvements in people's quality of life.